

Prosodic features of second language fluency

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Fluency has often been ignored in speech prosody research, including in research on prosody of a second language (L2). However, a broader view on L2 sentence prosody would be beneficial to the construction of theories concerning the acquisition of L2 prosody and applications such as assessments in teaching, exercises for individual learning, and automatic assessment of spoken performances. The contributions to this special issue give a good demonstration in which directions L2 speech prosody and fluency research should be extended. This concerns the integration of fluency, prosody, and gesture, a more detailed consideration of rhythm measures, and more local and dynamic perspectives on fluency. At the same time, conclusions from the combined contributions indicate other issues for future directions of investigation: thorough considerations of interactional fluency, clear definitions of and distinctions between disfluencies that help versus those that hinder communication, and how and whether human judgements can be used as point of reference in such research. Finally, while (large) between-speaker variations are known and under investigation in L2 fluency and prosody research, similar individual differences are yet to be understood when it concerns L1 speech. We argue for an integration of prosodic aspects of fluency in L2 teaching and assessment, for also taking perceptual perspectives on fluency, and for a new view on between-speaker variation.

Keywords: disfluency, fluency, prosody

1. Introduction

There is no doubt that pronunciation and fluency both play important roles in the speech of second or foreign language (L2) speakers. That said, at first glance, the *Journal of Second Language Pronunciation (JSLP)* — as a journal with *pronunciation* in its title — might not be the perfect fit for *fluency* research, which is often

subsumed under different linguistic sub-disciplines such as applied linguistics and pragmatics, as it is the case in the Common European Framework of Reference for Languages (CEFR) (Council of Europe 2020). But *JSLP* has a distinct focus on prosodic and segmental pronunciation of L2s and also provides connections between L2 speech and other areas of applied linguistics and pragmatics. Thus, it is the aim of this special issue to show links between segmental and suprasegmental aspects in speech fluency and to offer a broad and up-to-date perspective on speech fluency in L2 learning environments.

In the past, fluency has often been ignored in speech prosody research – as reflected, for instance, in the *Handbook of Language Prosody* (Gussenhoven & Chen 2020) and in the series of International Conference on Speech Prosody (e.g., Chen, Chen & Arvaniti 2024; Kuang & Liberman 2026). Moreover, fluency and timing are only rarely treated in conjunction with other prosodic features, in particular intonation, in L2 research. However, a broader view on L2 sentence prosody would be beneficial to the construction of theories concerning the acquisition of L2 prosody and applications such as assessments in teaching, exercises for individual learning, and automatic assessment of spoken performances. Likewise, research on language learning does not generally seem to be integrated into speech prosody research. This concerns both theoretical and methodological aspects but also acquisition and annotation of learner data, e.g., in learner corpora. Thus, the scope of this special issue includes topics like measuring fluency, assessment of fluency, learner data and annotation of disfluencies, elements and combinations of disfluencies (e.g., filler particles, disfluent pauses, lengthenings, repetitions, repairs), varying degrees of fluency in different speech styles and tasks, fluency and L2 proficiency levels, intonational but also visual aspects of fluency (e.g., handarm gestures).

This introduction to the special issue on “Prosodic features of second language fluency” in the *Journal of Second Language Pronunciation* is structured as follows. Section 2 is devoted to a definition of speech fluency, with a particular focus on L2-specific fluency. Section 3 provides short summaries of the four contributions to this special issue, which all widen specific aspects of current research on speech fluency. In the final section we conclude what we think we learned and in which avenues would be worth pursuing to further enhance our perspectives on prosody and its role in L2 fluency.

2. Defining speech fluency

Fluency refers to a behaviour that unfolds in time in which intended and expected movements follow a more or less fluent production. Fluency can be related to playing a musical instrument (including the voice) but also to walking, to

sequences in sports (e.g., running for a hurdle race), and also in cooking (e.g., preparing a menu). Often, such movements are organized in larger movement patterns that represent automated routines. There are units of movements that are intended in the process of execution and expected by recipients. A continuous execution of these units is usually regarded as fluent, whereas unintended and unexpected interruptions of this process are usually regarded as disfluent. Since humans are not entirely continuous in their intended movements all the time, the concept of fluency is better reflected by a gradient scale rather than a categorical distinction between fluent and disfluent.

Modality

Using language always implies timing and fluency, no matter in which modality. This holds for written language (hand-writing, machine-writing, silent reading, but also reading aloud), sign language (encoding and decoding sign language), and of course spoken language (speaking and listening). Although there are studies on fluency of writing (e.g., Wolf-Quinteiro et al. 1998) and fluency of signing (e.g., Marshall et al. 2018, and Haug et al. 2024, see also CEFR 2020, p.170), the vast majority of fluency research is on speech, and many of these studies have a focus on a second or foreign language, often in comparison to native speech (L1).

Typical speech disfluencies

No matter whether L1 or L2 speech is in focus, some types of disfluencies are mentioned more often than others, depending on the specific research question. In addition, the same or similar phenomena are often referred to by different labels, and sometimes events with the same label are used in different ways. Thus, there is no established standard of categorization and annotation of disfluencies in speech, as Trouvain et al. (2025) found in a study with 23 expert annotators from various research backgrounds who had to identify and label disfluencies in the same speech sample. Despite substantial variations between the annotators on various levels of annotation, the following coarse labels are frequently used: filler particles (fillers, filled pauses such as *uh* and *um* in English), lexical fillers (such as *y'know*, *like*, *so*, *I mean* in English), lengthenings (prolongations, drawls), overlong pauses, repetitions of syllables or words, false starts, and repairs (self-corrections). Lengthenings and overlong pauses are usually based on the perception of the annotator taking the phonetic context into account. Filler particles are best known as filled pauses (both can be abbreviated FPs); however, 'filled pause' is actually not the optimal label for the phenomenon of articulating non-lexical particles such as [ə:] and [ə:m], because it refers to the syllable-like articulated

section and not the entire pause (see Figure 1 top left), and sometimes there is no pause at all preceding and/or following the filled pause (see Figure 1 top right) (cf. Muhlack et al. 2023). The counterpart to the filled pause is the silent pause. The problem here is that many of the sections labeled as silent pauses are not silent at all (Belz & Trouvain 2019) (see Figure 1 bottom), many of them contain audible inhalation noises. It is often unclear when pauses without an FP should be considered as a disfluency, because pauses in speech are a natural phenomenon reflecting syntactic-prosodic phrasing structure. Yet, when pauses are overlong or in an unexpected position, they are usually regarded as hesitations.

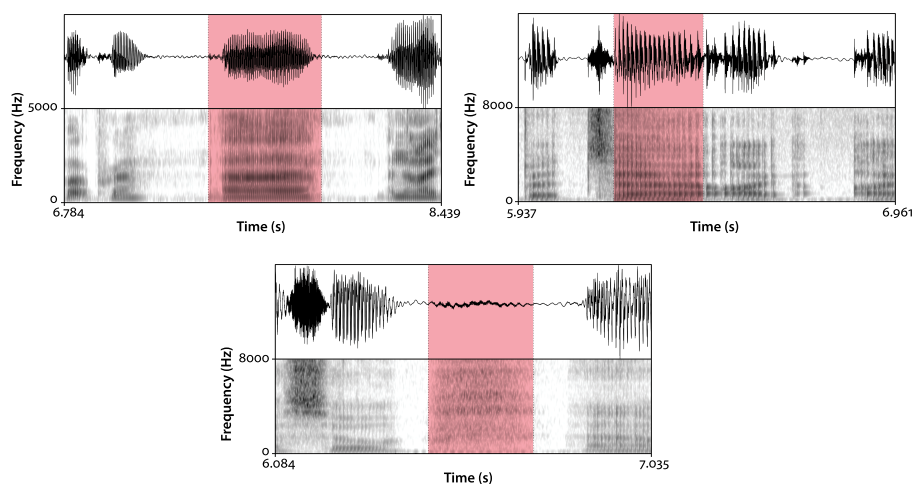


Figure 1. Speech signals with a time signal (top left) and a spectrogram with a display of 0-8 kHz (bottom)

Top Left: An example of an FP with silence: [ə:] flanked by a silence to the left and to the right. For some authors only the [ə:] is the FP, for others the entire pause that consists of silence plus [ə:] plus silence.

Top Right: An example of an FP without any silence: [ə:] without any surrounding in the middle of an articulation phase.

Bottom: An example of a silent pause that is not phonetically silent: an audible inhalation noise flanked by silence on the top left and the top right.

The often unspecified usage of FPs and (silent) pauses can lead to an undesired variability in measurement, e.g., when durations of these units are under study, but also in the treatment of FPs in those sections in which a speaker combines several disfluencies into one complex disfluent phase. In addition, some FPs seem to consist of glottal reflections, such as a creaky voiced *hm* (Belz, this volume), or a tongue click (Trouvain 2014), i.e., phonetic categories that have often been neglected. Moreover, most studies use the labels *uh* and *um*, which

doubtlessly exist in English, but it is not clear at all whether *uh* and *um* also exist in other languages, which may or may not use other types of FPs (see also Muhlack et al. 2023). As a last point on FPs, it should be mentioned that results of some studies suggest that disfluencies in read-aloud speech differ from those in spontaneous conversations; for instance, FPs rarely occur in read speech (e.g., Trouvain et al. 2026), in a stark contrast to many spontaneous speech productions. However, read speech is used for fluency assessment, especially in speech technology (e.g., Cucchiariini et al. 2000), and presumably also sometimes in language teaching in classrooms.

Fluency research outside L2 speech

Before going into more detail of the prosodic features of L1 and L2 speech fluency, it should also be mentioned that there are fields of speech fluency other than L2 that are under study but not considered here. Firstly, the acquisition of reading also involves fluency skills of young readers (e.g., Kuhn et al. 2010). Secondly, disorders of speech fluency such as stuttering and cluttering show patterns of interrupted speech flow in terms of unusual pauses (blocking), unusual lengthenings, and repeats of parts of a syllable (Lickley 2015, Crible et al. 2024, Didirková 2024). Such features of **dysfluency** can occur in addition to those features of **disfluency** (please note the spelling difference) that are common in the spontaneous speech of every L1 speaker (e.g., Shriberg 1994), including articulation rate, pauses, filler particles (e.g., *uh*), false starts, and repeats. It must be noted that several authors use the term dysfluency to refer to both meanings, viz a disorder of fluency skills and the interruption in non-disordered speech.

Another group of speakers with specific characteristics of speech fluency are patients with a neurodegenerative disease such as Alzheimer who produce many more and much longer pauses in verbal fluency tasks such as “name as many zoo animals as possible in the next 30 seconds” (Bucks et al. 2000), and are less fluent in spontaneous speaking tasks, than non-pathological speakers. Indeed, automated measures of fluency are used for identifying patients with Alzheimer’s dementia (Shah et al. 2021).

L2 fluency in speaking

This special issue, however, is focusing on online fluency in L2 speaking. In L2 research, “fluency” is one of the three main features in the CAF-framework to track development, the other two being “complexity” and “accuracy” (Housen & Kuiken, 2009). Throughout their development, learners will add new words and new grammatical structures to their repertoire, rendering their speech increasingly

complex (the “C” in CAF). Additionally, it takes time to learn how to use a (new) sound, lexical item, and syntactic structure accurately and contextually appropriately (“A”), leading to increasingly target-like production as proficiency develops. Finally, through massive and repeated use of known words and structures, speech will become more fluent (“F”), with faster speech rates and fewer unnatural hesitations.

From this brief description it can be hypothesized that the three main features are interconnected and develop together (see Vercelotti, 2017). At the same time, learning to speak another language is a dynamic process where learning to use a new and complex feature accurately and fluently without unnatural hesitations is unlikely to happen all at once. Indeed, trade-off effects have been hypothesized (e.g., Skehan, 1998) and found. Thai and Boers (2016), for instance, found a trade-off effect when L2 learners had to perform under time-pressure: fluency increased under this condition, but this came at the cost of decreased complexity and accuracy. In a longitudinal design, Yu and Lowie (2020) investigated development of accuracy and complexity and found a trade-off between accuracy and complexity at the start of the study by showing negative correlations between measures of the two aspects, but by ten weeks later, measures of accuracy and complexity supported each other.

In short, in the CAF framework, fluency is one of the indicators of overall speaking proficiency in a second language. This narrow notion is in contrast to the broader lay term of fluency referring to overall speaking proficiency (Lennon, 1990), for instance when “fluent” is used on a resume. The current special issue is concerned with the specialized, narrow notion. This narrow conceptualization of fluency can be seen from three different angles, as explained by Segalowitz (2010) and also taken by Lickley (2015). From a cognitive perspective, fluency in speaking refers to the cognitive operations carried out by the speaker: are the speaking processes that need to run in order to articulate the appropriate sounds that reflect the intended message smooth and efficient? Utterance fluency, in turn, is what is observable once speech is uttered and denotes the measurable aspects in the speech itself. Perceived fluency, finally, is how listeners perceive and judge the fluency of the speaker to be, based on what they hear from the utterance. So where cognitive fluency and perceived fluency are both invisible cognitive constructs, utterance fluency is the stuff that is on the minds of phoneticians and scholars concerned with prosody.

Typically, utterance fluency is subdivided into three aspects: breakdown, speed, and repair fluency (Tavakoli & Skehan, 2005). Whenever any of the speech production processes is halting, or running into trouble, a speaker may need to stop (“break down”) the current speech flow, measurable as FPs and overlong pauses. With respect to speed fluency, in an uninterrupted speech chunk, fast

speech production processes may lead to faster speech, measurable as a relatively high number of syllables per second (s/s). Speech tempo can either be reflected as speaking rate (s/s including pauses) or articulation rate (s/s excluding pauses). Finally, repairs in an utterance reflect situations where the speech production process resulted in overt or sometimes covert mistakes and thus cannot be considered smooth and efficient.

In L2 research, a wealth of studies has shown that objective measures of utterance fluency are related to measures of perceived fluency (e.g., Bosker et al., 2013; Kahng, 2014; Révész, Ekiert, & Torgersen, 2016) and to overall L2 speaking proficiency, even when rating scales used for L2 speaking proficiency do not explicitly include aspects of fluency (Kahng, 2014; Révész et al., 2016).

At the same time, it should be noted that a *high* degree of a respective utterance fluency type does not necessarily reflect ‘better’ speech. A too high speed of speaking reduces the level of intelligibility and the ease of comprehension (Munro & Derwing, 2001). Likewise, a *low* degree of a respective utterance fluency type does not necessarily have a negative impact on information transportation. For instance, the usage of FPs such as *um* and *uh* is beneficial for information processing, at least when listening to L1 speech. It has been shown that listeners raise their attention upon hearing disfluencies (Collard et al., 2008) and anticipate complex syntax (Watanabe et al., 2008) or low-probability words or low-frequency words (Bosker et al., 2014). Likewise, L1 disfluencies signal upcoming cognitively complex speech, such as new (Barr & Seyfeddinipur, 2010) or complex referents (Arnold et al., 2007).

In other words, some “disfluencies” help communication and may be better qualified as “fluencemes” (Götz, 2013). They are part of normal communication and signal upcoming difficult speech, be it at the conceptual, syntactic, or lexical level. This information is helpful for listeners to anticipate and better comprehend the incoming speech. At the same time, disfluencies that signal speech production processing difficulties that are not relevant to the listener, e.g., an L2 speaker searching for a high-frequency referent, a correct morphological ending, or article, will likely hinder effective communication and may thus indeed be called “disfluencies”, precisely because predictions by listeners on incoming speech are not borne out. The concept of “fluenceme” could also be applied to those FPs that are articulated without any silence and that are part of the articulation flow of a prosodic phrase (see Figure 1 middle). Such FPs do not hinder the flow of articulation, in contrast, they enable a continuation of the flow of articulation.

For the most part, however, different kinds of disfluencies/fluencemes have thus far mostly been treated similarly in L2 research and assessment resulting in a very general rule: the more disfluencies or the slower the speech, the less proficient. Recently, however, at least with respect to the syntactic location of pauses,

researchers have shown that not all pauses should be treated equally. Pauses between the largest syntactic units (spoken sentences or “AS-Units”, Foster et al., 2000) are natural and occur in L1 speech as often as in L2 speech (De Jong, 2016). Likewise, between clauses, pauses are more natural and perceived as more fluent than within clauses (Tavakoli, 2011; Kahng, 2018; Hsieh et al., 2019; Suzuki & Kormos, 2020).

Related to the issue that there are disfluencies/fluencemes that help communication by signaling upcoming difficult speech versus disfluencies that hinder communication, Segalowitz (2010) already pointed to the notion of L2-specific fluency. He argued researchers should separate disfluencies (and fluencemes) that may originate from L2-specific difficulties from those that originate from personal speaking style. A meta-analysis (Gao & Sun, 2024) summarized research investigating to what extent individual differences in L1 utterance fluency relates to L2 utterance fluency, showing that for most measures, extensive overlap indeed exists. Anecdotally, it is well known that people differ widely in the way that they speak: some speak slowly and others fast; some use more filled or silent pauses than others or use only few, and some may use longer pauses than others. Importantly, there is now empirical evidence showing that such individual differences carry over to how these people speak in their L2. If you are an “um-er”, you are an “um-er”, no matter which language you speak. For research into L2 speaking, whenever researchers are interested in L2 proficiency or L2-specific behavior, they should therefore take L1 behavior into account, for instance by residualizing L2-specific measures from L1-measures.

In many L2 speaking assessments, raters are asked to give judgements on levels of fluency. However, what listeners attend to is not always known. Listeners or raters are essentially a “black box” with potential unknown biases. Likewise, human judgments of comprehensibility, accentedness, and nativeness are prone to potentially unknown biases. Especially when the constructs that they need to attend to are ill-defined in the research, raters may follow internal ideas on what constitutes, for instance, comprehensibility or nativeness. In part, listeners in this case may base their judgement on the extent to which the voice sounds pleasant. When evaluating fluency, with precise definitions including known aspects of utterance fluency, however, it has been shown that naïve listeners indeed base their judgements on speed, breakdowns, and to a lesser extent on repairs (Bosker et al., 2013) and that they are able to tease apart fluency from another construct like pronunciation (Pinget et al., 2014).

Educational aims in learning, teaching, and assessment of L2 fluency

A less precisely defined concept of fluency can also have an impact on how teachers evaluate the spoken performances of their students. In most European countries, the CEFR has represented the standard platform for L2 teaching and assessment for decades. There are three big competence areas in CEFR (Council of Europe 2020): “pragmatic competence”, “sociolinguistic competence” and “linguistic competence”. Interestingly, the CEFR relates fluency only to “pragmatic competence” with no link at all to prosody. For the latter, a sub-area “phonological control” contains a sub-field on “prosodic features”.

Table 1. “Prosodic Features” (CEFR pp.134f.) and “Fluency Features” (CEFR p.141) of the beginner level (A1) and the most skilled level (C2) in CEFR

Level	Prosody	Fluency
A1	“Can use the prosodic features of a limited repertoire of simple words and phrases intelligibly, in spite of a very strong influence on stress, rhythm and/or intonation from the other language(s) they speak; their interlocutor needs to be collaborative.”	“Can manage very short, isolated, mainly pre-packaged utterances, with much pausing to search for expressions, to articulate less familiar words/signs, and to repair communication.”
C2	“Can exploit prosodic features (e.g. stress, rhythm and intonation) appropriately and effectively in order to convey finer shades of meaning (e.g. to differentiate and emphasise).”	“Can express themselves at length with a natural, effortless, unhesitating flow. Pauses only to reflect on precisely the right means to express their thoughts or to find an appropriate example or explanation.”

So, in CEFR, fluency is considered from speech planning and pragmatic perspectives only, and then strictly separated from prosody as such, though fluency can be often operationalised in detail in prosodic means. There are doubts that such a strict separation is useful because it closes the eyes for the clear link between fluency and prosody.

Further fluency challenges for L2 learners

So far, the production aspects of the learners of an L2 were in focus. For perception, L2 learners face a special challenge, because L1 speech is characterised by more fluent phases than L2 speech. This means for an L2 learner that when listening to L1 speech, processing must cope with fewer and shorter pauses, faster articulation, and many phonetic reductions (Ernestus et al. 2017). Thus, fluency

while listening presents special challenges for L2 listeners, where a distinction into segmental vs. suprasegmental phenomena is obsolete. Speech pauses represent a further important type of fluency-related feature relevant for both speaking and listening. Overall, the prosodic parameters of articulation rate, number and duration of pauses, including duration of inter-pause intervals, have a strong impact on the processing of speech in terms of production *and* perception.

3. Contributions to this special issue

The papers in this special issue mainly contribute to two topics introduced so far. Firstly, fluency and prosody must not be studied separately (Van Maastricht), for instance because the prosodic aspect of rhythm in L2 speech affects listener perceptions along with aspects of fluency (Fraser, Mora, & Ortega). The second main topic addressed is related to the discussion on disfluencies or fluencemes. Coulange and De Jong as well as Belz contribute to our understanding that the specific use and location of (filled) pauses matter. A simple count of such hesitations, concluding that higher equals less fluent, does not suffice.

3.1 Van Maastricht

The paper “*Fluency, prosody, and gesture: Interactions in L2 production and L1 perception*” by Lieke van Maastricht synthesizes research on how fluency, prosody, and gesture interact in L2 speech, integrating domains that are often studied separately. A central issue addressed is the persistent difficulty L2 learners face in acquiring prosody — features such as rhythm, stress, and intonation. These challenges are largely attributed to cross-linguistic transfer, where learners apply L1 prosodic patterns to the L2, resulting in deviations that affect both production and perception. Such deviations often lead to increased perceived foreign accent and reduced comprehensibility, even at advanced proficiency levels.

The paper conceptualizes fluency as a multidimensional construct, distinguishing between cognitive fluency (processing efficiency), utterance fluency (observable features like pauses and speech rate), and perceived fluency (listener judgments). Importantly, many utterance-level indicators — such as pausing, timing, and rhythm — are inherently prosodic, highlighting the overlap between fluency and prosody. The review then examines prosody in L2 production, defining it as suprasegmental features realized through acoustic cues like pitch, duration, and intensity. Because languages differ in how these features are structured, learners often struggle to approximate target-like prosody, particularly when L1 and L2 differ typologically.

A key contribution is the analysis of how prosody and fluency affect L1 listener perceptions, especially across four dimensions: accentedness, nativeness, comprehensibility, and intelligibility. Prosodic inaccuracies strongly influence accentedness, while comprehensibility is more closely linked to fluency-related measures such as speech rate and pausing. Perceived fluency correlates closely with comprehensibility but less with accentedness, indicating that ease of understanding and native-likeness are distinct perceptual constructs. The paper further argues that fluency and prosody should not be treated as independent domains. Instead, fluency measures can be seen as part of the prosodic system, which helps explain their combined impact on listener judgments.

Finally, the review introduces gesture as a multimodal component of L2 communication. Research suggests that gestures – such as rhythmic “beat” gestures – can support fluency and improve prosodic production, while other gestures may reflect cognitive effort and correlate with disfluency. However, findings are mixed, and the effectiveness of gestures depends on factors such as task type, learner characteristics, and gesture form.

In conclusion, the paper emphasizes that fluency, prosody, and gesture are tightly interconnected in L2 speech. It calls for more integrative research and suggests that language teaching should give greater attention to prosody and potentially incorporate gesture-based methods to enhance communicative effectiveness.

3.2 Coulange & De Jong

The paper “*Fluency assessment: incorporating syntactic distance in a new measure of pause location*” by Sylvain Coulange and Nivja de Jong proposes a novel approach to assessing L2 speech fluency by focusing not only on how often and how long speakers pause, but crucially where pauses occur relative to syntactic structure. Traditional fluency measures – such as articulation rate, pause frequency, and pause duration – primarily treat pauses as disruptions. However, the authors argue that pauses can also be functional, especially when they align with meaningful syntactic boundaries, aiding speech planning and listener comprehension.

Previous research has shown that more proficient speakers tend to pause at higher-level boundaries (e.g., between clauses), while less proficient speakers pause more frequently and in less appropriate locations. Most existing approaches model pause placement using categorical distinctions (e.g., clause-internal vs. clause-external pauses), but these are too coarse to capture the gradient nature of syntactic structure. To address this limitation, the authors introduce a continuous measure of syntactic distance, defined as the number of syntactic constituents

opening or closing between adjacent words. This allows pause locations to be quantified on a gradient scale reflecting syntactic complexity. Building on this, they propose the Syntactic Pause Ratio (SPR), which relates pause placement to the underlying syntactic structure of speech.

The study analyzes spontaneous English speech from Japanese learners (B1–C1 levels) and native speakers, using automated transcription and syntactic parsing. Statistical models are used to examine whether syntactic distance predicts pause behavior and whether the new metric improves proficiency assessment. Results show that syntactic distance strongly predicts pause occurrence: speakers are more likely to pause at points of greater syntactic complexity. This continuous measure outperforms categorical approaches in predicting where pauses occur. Native speakers demonstrate greater sensitivity to syntactic structure, placing pauses more strategically than learners. For pause duration, both syntactic distance and categorical boundary types are predictive, though neither clearly outperforms the other. Lower-proficiency speakers tend to produce longer pauses at complex boundaries, likely reflecting increased processing difficulty. Most notably, the inclusion of SPR substantially improves proficiency prediction. While traditional fluency measures explain a moderate proportion of variance, adding syntactic information increases explanatory power to around 84–87%, highlighting the importance of pause placement relative to syntax.

The study concludes that fluency is not simply about speaking faster or pausing less, but about pausing appropriately within syntactic structure. By introducing a continuous syntactic framework, it offers a more nuanced and powerful model of L2 fluency, with implications for both automated assessment and language teaching.

3.3 Fraser, Mora & Ortega

The paper “*Examining the relationship between speech rhythm, fluency measures and comprehensibility ratings in L2 English speech*” by Katherine Fraser, Joan Carles Mora, and Mireia Ortega examines how speech rhythm and fluency interact in L2 English and how both contribute to listeners’ perceptions of comprehensibility and fluency. It addresses a key gap by jointly analyzing these constructs in spontaneous speech rather than treating them separately.

Speech rhythm is defined as the temporal organization of stressed and unstressed syllables, often associated with durational variability across segments. Traditional rhythm metrics (e.g., %V, nPVI-V) attempt to quantify this variability, but they have been criticized for their indirect connection to phonological structure. As an alternative, the authors emphasize vowel reduction — differences in duration between stressed and unstressed vowels — as a more direct and linguistics-

tically meaningful indicator of English rhythm. Fluency is conceptualized within a standard framework that distinguishes speed fluency (speech rate), breakdown fluency (pauses), and repair fluency (repetitions and self-corrections). These dimensions reflect processing efficiency and temporal smoothness in speech production and are known to affect listener judgments.

The study analyzes speech from 82 advanced Spanish–Catalan learners of English performing a spontaneous task. The researchers extracted multiple rhythm and fluency measures and obtained ratings of comprehensibility and perceived fluency from native English listeners. Results show that rhythm and fluency are only partially correlated, indicating they capture distinct aspects of L2 speech. While faster speech was sometimes associated with more target-like rhythmic variability, many traditional rhythm metrics showed weak or inconsistent relationships with fluency measures. This supports the view that rhythm is not reducible to fluency. Crucially, both domains independently predicted listener ratings. Speech rate and pausing were strong predictors, confirming prior research. However, rhythm – especially vowel reduction – also significantly influenced comprehensibility and perceived fluency. In fact, measures of unstressed vowel duration were among the most consistent predictors, whereas traditional variability metrics contributed less. Repair fluency had little effect, suggesting that listeners are less sensitive to such features.

The study proposes that rhythm and fluency are complementary components of L2 speech performance. Fluency reflects processing efficiency, while rhythm reflects phonological timing patterns. Both are necessary to explain listener perceptions. Pedagogically, the findings highlight the importance of rhythm – especially vowel reduction – as a target for instruction alongside fluency. However, developing target-like rhythm may depend on achieving sufficient fluency. The authors note limitations, specifically the focus on one learner group and on durational measures, and call for broader, multidimensional research on L2 prosody.

3.4 Belz

Malte Belz's paper, "*Measuring filler particles in speech fluency: Static and dynamic analyses*," examines how FPs such as *uh* and *um* should be conceptualized and measured in L2 fluency research. Although FPs are commonly treated as indicators of disfluency, the paper argues that their role is more complex and requires both conceptual and methodological refinement (see discussion in Section 2). The study begins by highlighting a key problem: FPs do not fit neatly into established fluency categories such as speed, breakdown (pauses), or repair. Depending on context, they can signal hesitation and planning difficulty, but they can also facilitate communication by structuring discourse or managing turn-taking.

Because of this dual function, Belz proposes treating them as a separate domain, termed filler particle fluency, rather than grouping them with pauses or repairs.

A central critique concerns the widespread use of static measurement approaches, such as counting FPs per number of words or per minute. While useful for comparison, these aggregate metrics obscure how FP usage varies over time. They fail to capture whether FPs occur sporadically or cluster in moments of increased cognitive load, which limits their explanatory power. To address this, the paper introduces dynamic, time-sensitive methods that treat fluency as a process unfolding over time. Using dialogue data from 24 speakers, the study applies several techniques: section-wise analysis, cumulative frequency tracking, and especially sliding window analysis. The latter involves calculating FP frequency across overlapping segments of speech, producing a continuous trajectory of FP usage. This approach reveals local fluctuations – periods of high and low FP density – that static averages cannot detect.

The findings show that FP usage is highly variable and context-dependent. Rather than being evenly distributed, FPs tend to cluster around cognitively demanding moments, supporting the idea that fluency is dynamic rather than a stable speaker trait. However, the paper also notes challenges in interpreting these patterns. Peaks in FP frequency may reflect hesitation, discourse organization, or interactional strategies, making functional interpretation complex. The study further emphasizes the importance of clear definitions and consistent annotation practices, as ambiguity in what counts as an FP undermines comparability across studies.

In conclusion, Belz argues for a more nuanced understanding of fluency that recognizes filler particles as a distinct and dynamic phenomenon. That said, he also notes that the study of FPs should ideally involve other fluency features such as pauses and repairs. Finally, he suggests that adopting time-series methods might facilitate capturing the temporal structure of speech.

4. Conclusions and implications

4.1 What have we learned and where do we go from here?

Integrating fluency, prosody, and gesture

Van Maastricht points to a relatively new line of research that promises to lead to a new understanding by combining different related research fields. Some studies have already shown that L2 learners produce more gestures than L1 speakers do (e.g., Gullberg, 1998; Nagpal et al., 2011), possibly to cope with the cognitive difficulties they experience while speaking in an L2, compared to an L1 (e.g., Peltonen

et al., 2023; Lopez-Ozieblo, 2022). A relatively new area of research is combining both fluency and gesture phenomena and observing how they co-occur in both L1 and L2 speech. This line of research has found that gestures tend to co-occur with fluent rather than with disfluent speech (Peltonen et al., 2023; Lopez-Ozieblo, 2022; Kosmala, 2021; Graziano & Gullberg, 2018).

Interactional fluency

Studies in this special issue investigated fluency in monologues (Fraser, Mora, & Ortega), dialogues (Belz), and in ‘long’ turns (at least 8 seconds) in dialogues (Coulange & De Jong). Speaking modes have repercussions on how disfluencies and fluencemes are manifested (Tavakoli, 2016). Although an early seminal study by Riggensbach (1991) discussed how measures for monologic fluency differ from those for interactive fluency (labeled “conversational fluency”), most L2 research since then has focused on monologues or long turns in a simulated dialogic setting. More recently, L2 researchers have noted the importance of Interactional Fluency (IF) as a separate construct. Peltonen (2020) defined IF as a collaborative aspect of fluency, that is apparent in between turns and results in a smooth conversational flow. IF thus requires coordination from speakers and listeners across turn boundaries and being fluent in interaction means that not only speaking processes but also listening processes must run smoothly and efficiently. Because gaps between turns in a fluent conversation usually do not take longer than 200 milliseconds (Stivers et al., 2009), speakers need to prepare their next turn while listening to, processing, and comprehending the interlocutor’s message. Thus, fluent turn-taking incorporates prediction during a conversation. L1 speakers indeed effectively predict what their interlocutors will say next and when the current turn will end so that they can take the next turn in time. In contrast, for L2 speakers turn-taking represents two major obstacles for IF: they have to tune their perception for an appropriate prediction of the turn-taking opportunity and they have to take care of planning and producing their own turn. Pickering and Gambi (2018) describe how the predictive processes are less efficient for L2 speakers, leading to predictions that are less precise compared to those from L1 speakers (see also Sbranna 2025). A novel promising line of research combines IF with research on gestures. In this line of research, Lin et al. (2026) show that high-proficient L2 speakers manage to coordinate their turn transitions with gestures, supporting smoother turns.

Between-speaker variation

Fraser, Mora, and Ortega mentioned that between-speakers differences were found to be larger than between-language differences, and the figures in Belz’s and Coulange & De Jong’s studies clearly show large individual differences as well

as overlap between L1 and L2 speech measures. As mentioned in Section 2, such large individual differences do not necessarily relate to L2 proficiency, as some individual differences may carry over from L1 speech. One line of research may consider such L1 individual differences when investigating L2 speech (as mentioned in Section 2); another, more fundamental line of research should investigate how L1 individual differences can be explained, not only with respect to fluency but with respect to other linguistic behavior as well. In this line of research, for instance, Llompart & Dabrowska (2023) show that language aptitude, a construct usually employed in L2 research, also predicts L1 grammatical proficiency.

Dynamic and local fluency

Operationalization of dynamic fluency (as in Belz) and operationalizations of local fluency (as in Belz and Coulange & De Jong) may be other future avenues in research on L2 fluency. Fluency is not only a dynamic notion from week to week as it may progress over time along (or in competition) with accuracy and complexity, but also dynamic on a much smaller time scale. Whenever the upcoming part of an utterance is at a major syntactic break with the previous part of the utterance, speakers are more likely to pause (Coulange & De Jong). To gauge how fluency develops over time during the time course of one single monologue or dialogue, more information can and should be added in future research. A number of such potential factors have been studied earlier in combination with fluency, for instance word frequency (De Jong, 2016), emotion (Aubrey, 2022), and discourse information such as given versus new referents or topic change in L1 speech (Barr & Seyfeddinipur, 2010; Swerts, 1998). None of the research so far, however, has managed to study these factors in conjunction, to truly track underlying cognitive and emotional causes of disfluencies in real time.

Communicative adequacy

In Section 2, we introduced fluency as one of the three linguistic aspects in which L2 speakers may develop, alongside complexity and accuracy. However, this trio of constructs is not sufficient to evaluate speaking performance. L2 speakers may use (appropriate) complexity in their utterances, be perfectly target-like and fluent while at the same time not reaching the intended communicative goal. This would happen, for instance, when an L2 speaker recites a previously practiced speech that is off-topic. Therefore, communicative adequacy or functional adequacy (Palotti, 2009) is a fourth relevant construct that needs to be considered when tracking development. Has the intended communicative goal been reached? If not (at all), then it may be irrelevant to evaluate the linguistic aspects such as fluency. For future studies into fluency and prosody of L2 speech, instead of only

focusing on human evaluations of intelligibility, comprehensibility, or accentedness, it may be worthwhile to (also) consider communicative adequacy as a relevant construct. Whichever construct is chosen, it is imperative that the listeners who are asked to rate the performances have been sufficiently instructed.

4.2 Implications

L2 fluency and L2 prosody research usually have a common goal in mind: to understand individual differences in learning trajectories and – ultimately – to help learners reach target-like and communicatively effective goals. The papers in the current issue may not all directly point to such pedagogical consequences, but such effects can still be discerned. Van Maastricht shows that hand gestures and other forms of embodiment may play a beneficial role when teaching L2 pronunciation. Fraser, Mora, & Ortega add that training speech rhythm is a promising avenue for teaching pronunciation. The studies by Belz and Coulange & De Jong clearly point out that with respect to fluency, a target-like goal does not entail speech without any pauses; rather some disfluencies (fluencemes) will help communication, and some may hinder effective communication. Teachers could show in class that proficient speakers also use pauses. This may help L2 learners recognize a more realistic goal in their attempts to develop their L2 speaking fluency. They may even show at which locations pauses are helpful or hinder, raising awareness of how discourse is organized. Finally, the pronunciation of filler particles like *uh* and *uhm* differs across languages. When teachers help learners become aware of these differences between their L1 and the target language, they will help learners to become fluently disfluent.

Regarding L2 assessment, implications are likewise not always at the forefront but can still be inferred from the findings. One implication for L2 speaking assessment is that not all disfluencies should automatically be interpreted as indicators of lower proficiency. The studies suggest that certain disfluencies may contribute to effective speech production and communication. Through time (Belz) and at specific syntactic locations (Coulange & De Jong), disfluencies help listeners predict upcoming complex speech and can thus be considered fluent pauses. Another implication follows from van Maastricht: gestures are part of communication and must therefore also be part of assessment. This point was already made by Plough et al. (2018), who indeed called for research on the link between interactional competence and nonverbal behavior.

In the first two sections we showed that there is not one homogenous community of speech fluency research, there are rather several communities that often do not share terminology and methods, but where a mutual exchange might be beneficial. As a prime example, research on speech prosody, particularly L2 prosody,

could consider L2 speech fluency research as part of prosody. All contributions in this special issue have shown this: Maastricht towards visual prosody, Fraser, Mora & Ortega towards rhythm, Coulangue & de Jong towards syntactic-prosodic phrasing, and Belz to discourse structure in dialogues. Language teaching and assessment, as reflected by the CEFR guidelines, could benefit when prosody and fluency are thought together. This would lead to exercises where prosodic properties such as pauses, articulation rate and disfluencies are practiced with specific pragmatic functions – both for production and perception. This would represent a better foundation for the training of interactional fluency. Research communities interested in language fluency outside L2 speech on the one hand and L2 speech fluency research on the other should engage in a mutual exchange of ideas. So far, L2 aspects of fluency in sign language users, in reading and writing, and in fluency-disordered speech seem to be treated in separate communities with their own journals and conferences but without a link to the others. Finally, L2 research in applied linguistics could be linked more closely to prosody research, especially when it comes to technological solutions for automatic and semi-automatic procedures of fluency assessment.

Funding

Open Access publication of this article was funded through a Transformative Agreement with Leiden University.

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Publication history

Date received: 6 July 2026
Date accepted: 9 July 2026
Published online: 18 September 2026